

Lec 26.

Recap.

Bayes rule

Given a prior and likelihood distribution can find the posterior distribution

$f_X(x)$, $P_X(x)$: Prior distribution

$f_{Y|X}(y|x)$ $P_{Y|X}(y|x)$
likelihood.

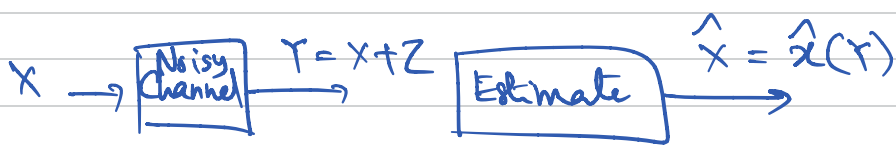
$P_{X|Y}(x|y)$, $f_{X|Y}(x|y)$.
Posterior probability

① X, Y are continuous

$$f_{X|Y}(x|y) =$$

$$\frac{\overbrace{f_{Y|X}(y|x) f_X(x)}^{f_{X,Y}(x,y)}}{\underbrace{\int_{-\infty}^{\infty} f_{Y|X}(y|u) f_X(u) du}_{f_Y(y)}}$$

② X is discrete & Y continuous.



$$P(X \neq \hat{x})$$

$$\hat{x}_{\text{sign}}(y) = \begin{cases} +1 & y > 0 \\ -1 & y < 0 \end{cases}$$

Example:

$X \sim \text{Uniform}(\{+1, -1\})$

$Y = X + Z$

$Z \sim N(0, \sigma^2)$.
 Z is independent X .

$\begin{cases} Y|X=+1 \sim N(1, \sigma^2) \\ Y|X=-1 \sim N(-1, \sigma^2) \end{cases}$

MAP estimator

(Maximum A Posterior Distribution).

$$\hat{x}_{\text{MAP}}(y) = \arg \max_{x \in \{+1, -1\}} P_{x|Y}(x|y)$$

$$= \begin{cases} +1 & P_{x|Y}(+1|y) \geq P_{x|Y}(-1|y) \\ -1 & P_{x|Y}(+1|y) < P_{x|Y}(-1|y) \end{cases}$$

$\hat{x}_{\text{MAP}}$ is same as $\hat{x}_{\text{sign}}$ estimator.
 $\forall x \in \{+1, -1\}$

$$P_{x|Y}(x|y) = \frac{f_{Y|X}(y|x) P_X(x)}{f_Y(y)}$$

$$\max_{x \in \{+1, -1\}} P_{x|Y}(x|y) = \frac{1}{f_Y(y)} \max_{x \in \{+1, -1\}} f_{Y|X}(y|x) P_X(x)$$

$$\arg \max_{x \in \{+1, -1\}} P_{x|Y}(x|y) = \arg \max_{x \in \{+1, -1\}} f_{Y|X}(y|x) P_X(x).$$
$$= \arg \max_{x \in \{+1, -1\}} e^{-\left(\frac{y-x}{\sigma}\right)^2 \frac{1}{2}} \frac{1}{2}$$

$$Y|X=1 \sim N(1, \sigma^2).$$

$$Y|X=-1 \sim N(-1, \sigma^2).$$

$$f_{Y|X=1}(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\left(\frac{y-1}{\sigma}\right)^2 \frac{1}{2}}.$$

$$f_{Y|X=-1}(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\left(\frac{y+1}{\sigma}\right)^2 \frac{1}{2}}$$

$$= \begin{cases} +1 & \frac{1}{2} e^{-\left(\frac{y-1}{\sigma}\right)^2 \frac{1}{2}} > \frac{1}{2} e^{-\left(\frac{y+1}{\sigma}\right)^2 \frac{1}{2}} \\ -1 & \text{otherwise} \end{cases}$$
$$-\left(\frac{y-1}{\sigma}\right)^2 \frac{1}{2} > -\left(\frac{y+1}{\sigma}\right)^2 \frac{1}{2}$$

$$\arg \max_x f(x) = x^*$$

Let $c > 0$.

$$\arg \max_x c f(x) = x^*$$

$$\left(\frac{y+1}{\sigma}\right)^2 - \left(\frac{y-1}{\sigma}\right)^2 > 0.$$

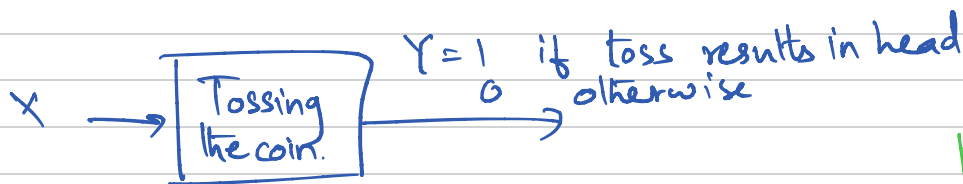
$$4 \frac{y}{\sigma^2} > 0.$$

$$y > 0.$$

$$\hat{x}_{\text{MAP}}(y) = \begin{cases} +1 & y > 0 \\ -1 & y < 0 \end{cases}$$

© X is continuous and Y is discrete.

$X \sim \text{Uniform}[0, 1] \rightarrow \text{Bias of a coin.}$



$$Y|X=x \sim \text{Bernoulli}(x).$$

$$f_X(x) = \begin{cases} 1 & x \in [0, 1] \\ 0 & \text{otherwise} \end{cases}$$

$$P_{Y|X=x}(y) = \begin{cases} x & y=1 \\ 1-x & y=0 \end{cases} = x^y \times (1-x)^{1-y}, \quad y \in \{0, 1\}$$

$$Y|P=p \sim \text{Bernoulli}(p)$$

$$P_{Y|P=p}(y) = \begin{cases} p & y=1 \\ 1-p & y=0 \end{cases}$$

$$f_{X|Y}(x|y) = \frac{f_X(x) P_{Y|X}(y|x)}{\int f_X(u) P_{Y|X}(y|u) du}$$

$$= \frac{1 \cdot x^y (1-x)^{1-y}}{\int_0^1 u^y (1-u)^{1-y} du}, \quad x \in [0, 1]$$

Suppose we see a head after the toss
 $\hat{u}_1, y=1.$

$$f_{X|Y}(x|1) = \frac{x}{\int_0^1 u \, du} = 2x \quad x \in [0,1].$$

Conditional Expectation

$\nearrow$ $E[X|Y=y]$ $\xrightarrow{\text{functions of } y}$ $\int_{-\infty}^{\infty} x \cdot f_{X|Y}(x|y) \, dx.$
 X is continuous.

$$E[X|Y=y] = \sum_{x \in \mathcal{X}_X} x \cdot P_{X|Y}(x|y)$$

X discrete

$$h(y) = E[X|Y=y].$$

Then $h(Y)$ is a random variable that is
a function of r.v. Y .

$$E[X|Y] = h(Y).$$

Total law of expectation

$$E[X] = E[E[X|Y]]$$

X, Y continuous.

We did the
proof earlier
for X, Y
discrete.

RHS $\therefore h(y) = E[X|Y=y]$.

$$E[E[X|Y]] = E[h(Y)] = \int_{-\infty}^{\infty} h(y) f_Y(y) dy.$$

$$= \int_{-\infty}^{\infty} \underbrace{E[X|Y=y]} f_Y(y) dy.$$

$$= \int_{-\infty}^{\infty} f_Y(y) \left[\int_{-\infty}^{\infty} \underbrace{x f_{X|Y}(x|y)} dy \right] dy.$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x \underbrace{f_{X|Y}(x|y) f_Y(y)} dx dy$$

$$= \int_{-\infty}^{\infty} x \int_{-\infty}^{\infty} \underbrace{f_{X,Y}(x,y)} dy dx.$$

$$= \int_{-\infty}^{\infty} x f_X(x) dx = E[X].$$

X continuous, Y discrete.

$$h(y) = E[X|Y=y] = \int_{-\infty}^{\infty} x \cdot f_{X|Y}(x|y) dx$$

$$E[X|Y] = h(Y).$$

$$E[E[X|Y]] = E[h(Y)]$$

$$= \sum_{y \in \mathcal{Y}} P_Y(y) h(y)$$

$$= \int_{-\infty}^{\infty} \sum_{y \in \mathcal{Y}} P_Y(y) x f_{X|Y}(x|y) dx$$

$A_1, A_2, \dots$ partition $\mathcal{A}$

$$f_X(x) = \sum_i P(A_i) f_{X|A_i}(x)$$

$$A_y = \{Y=y\}$$

$$= \int_{-\infty}^{\infty} x \left[\sum_{y \in \mathcal{Y}} P_Y(y) f_{X|Y}(x|y) \right] dx$$

$\parallel$
 $f_X(x)$

$$= \int_{-\infty}^{\infty} x f_X(x) dx$$

$$= E[X].$$

Continuing the biased coin toss example.

$X \sim \text{Uniform}[0, 1]$.

$$f_{X|Y}(x|i) = 2x \quad x \in [0, 1].$$

To show that

$$E[X|Y=0] \leq E[X] \leq E[X|Y=1].$$

Exercise.

$$E[X|Y=1] = \int_0^1 x f_{X|Y}(x|1) dx.$$

$$= \int_0^1 x \cdot 2x dx.$$

$$= 2 \left. \frac{x^3}{3} \right|_0^1 = 2/3 = 0.67.$$

$$E[X] = \int_0^1 x \cdot f_X(x) dx = \int_0^1 x dx$$

$$= \left. \frac{x^2}{2} \right|_0^1 = 1/2 = 0.5.$$

Functions of random variable.

For discrete R.V.s X, Y st $Y = g(X)$ and given PMF of X , $P_X(x)$.

$$P_Y(y) = \sum_{x: g(x)=y} P_X(x)$$

For continuous r.v.s X and $Y = g(X)$.

given pdf of X , how to find pdf of Y .

- Find $F_Y(y)$ CDF of Y , using CDF / PDF of X .
- Differentiate $F_Y(y)$ to find the pdf $f_Y(y)$.

Examples ① $Y = \sqrt{X}$ where $X \sim \text{Uniform}[0, 1]$.

$$F_Y(y) = P(Y \leq y) = P(\sqrt{X} \leq y)$$

$$= P(X \leq y^2)$$

$$= F_X(y^2)$$

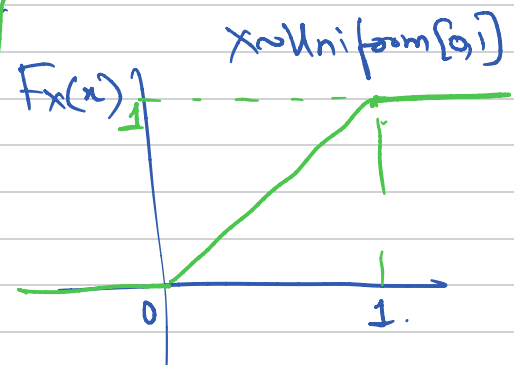
$$f_Y(y) = \frac{\partial F_Y(y)}{\partial y}$$

$$= \frac{\partial F_X(y^2)}{\partial y}$$

$$= f_X(y^2) \cdot 2y$$

$$\frac{df(g(t))}{dt}$$

$$= f'(g(t)) \cdot g'(t)$$



$$F_X(x) = \int_{-\infty}^x f_X(x) dx$$

$$= \begin{cases} \int_0^x 1 dx & x \geq 0 \\ 0 & \text{other} \end{cases}$$

$X \sim \text{Uniform}[0, 1]$

$$f_Y(y) = \begin{cases} 2y & 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$0 \leq y \leq 1$
otherwise

Exercise

$X \sim \text{Uniform}([0, 1]) \rightarrow$ Bias of coin # of heads in
 $Y|X=x \sim \text{Binomial}(n, x) \rightarrow$ n coin tosses

Find $f_{X|Y}(x|y)$ for $y \in \{0, 1, 2, \dots, n\}$.
 $x \in [0, 1]$.